

AI-Based Disease Prediction System

Using Machine Learning

Major Project Report

Submitted by: **Your Name**

Technology: Python • Pandas • NumPy • Scikit-learn

Model: Decision Tree Classifier

1. Problem Statement

Early prediction of diseases plays a critical role in timely diagnosis and treatment. Many patients delay seeking medical help because they are unsure whether their symptoms are serious. This project builds a machine learning application that predicts the most likely disease based on the symptoms entered by the user, encouraging early medical consultation.

2. Objectives

- Load a symptom–disease dataset.
- Perform data preprocessing.
- Train a machine learning classification model (Decision Tree).
- Accept symptoms from the user.
- Predict the most likely disease.
- Display the prediction in a clear, user-friendly format.
- Handle invalid or missing input gracefully.

3. Tools & Technologies

Tool	Purpose
Python 3	Core programming language
Pandas	Loading and preprocessing the dataset
NumPy	Numerical operations
Scikit-learn	Decision Tree model, train/test split, metrics
Joblib	Saving and loading the trained model (optional)

4. System Architecture

The system follows a simple, linear machine learning pipeline. Each stage feeds its output into the next:

Dataset (CSV) → Load (Pandas) → Preprocess (X / y split) → Train Decision Tree → User Symptom Input → Feature Vector → Prediction → Display Result

5. Dataset Description

The dataset (**disease_dataset.csv**) uses a binary symptom matrix. There are **20 symptom columns** (features) where 1 means the symptom is present and 0 means absent, and one **disease** column (target). It contains **10 diseases** with 60 samples each (600 rows total), including small realistic noise so the model learns robust patterns.

- **Symptoms:** fever, cough, fatigue, headache, sore_throat, runny_nose, body_ache, chills, nausea, vomiting, diarrhea, abdominal_pain, loss_of_taste, shortness_of_breath, joint_pain, skin_rash, dizziness, chest_pain, sneezing, sweating.

- **Diseases:** Common Cold, Influenza (Flu), COVID-19, Malaria, Dengue, Typhoid, Gastroenteritis, Migraine, Pneumonia, Food Poisoning.

6. Data Preprocessing

Preprocessing prepares the raw CSV for the model. Fully empty rows are dropped, any missing symptom values are filled with 0 (symptom absent), and all symptom columns are cast to integers. The data is then split into the feature matrix X (symptoms) and target vector y (disease).

```
X = df[symptom_columns]
y = df['disease']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42, stratify=y)
```

7. Machine Learning Model

A **Decision Tree Classifier** (scikit-learn) is used as the primary model. A decision tree learns a series of yes/no questions on the symptoms (e.g. 'Is fever present?') and follows the branches to reach a predicted disease at a leaf node. It is easy to interpret, fast to train, and well suited to binary symptom features. The model uses the 'entropy' criterion for information-gain based splits and achieves about **95% accuracy** on the held-out test set.

```
model = DecisionTreeClassifier(criterion='entropy', random_state=42)
model.fit(X_train, y_train)
```

8. Code Explanation

- **load_dataset()** — reads the CSV with Pandas and validates it exists.
- **preprocess_data()** — cleans data and splits it into X (features) and y (target).
- **train_model()** — trains the Decision Tree and reports test accuracy; optionally saves the model with joblib.
- **parse_user_input()** — converts comma-separated symptom numbers/names into a binary feature vector and warns about invalid entries.
- **predict_disease()** — runs the model and returns the predicted disease plus confidence scores.
- **display_result()** — prints the prediction in a clear, formatted box.
- **interactive_mode()** — repeatedly accepts symptoms until the user quits.

9. Screenshots

```
disease_prediction.py

disease_prediction.py
*****
WELCOME PREDICTION SYSTEM (Decision Tree)
*****
Loaded successfully: 600 rows, 21 columns.
Loading complete: 20 symptom features.
Model trained successfully. Test accuracy: 95.00%

-----
1. Fever      3. Fatigue
2. Sore Throat 6. Runny Nose
3. Chills     9. Nausea
4. Diarrhea   12. Abdominal Pain
5. ShortnessOf Breath ...

-----
```

Figure 1: Loading the dataset, training the model, and the symptom menu.

```
disease_prediction.py

Fever, cough, loss_of_taste, shortness_of_breath

=====
PREDICTION RESULT
=====
Fever, Cough, Loss Of Taste, Shortness Of Breath
>>> COVID-19 <<<

-----
Confidence):
100.0% #####

-----
ML-based prediction for educational
Please consult a qualified doctor.
=====
```

Figure 2: A disease prediction with confidence score.

```
disease_prediction.py

, 999
Unrecognised entries xyz, 999
The entered symptoms were valid. Please try again.

Headache, nausea, dizziness

Input: Headache, Nausea, Dizziness
>>> Migraine <<<
100.0% #####
```

Figure 3: Graceful handling of invalid input.

10. Results

The Decision Tree model achieved approximately **95% accuracy** on the test set and correctly classified canonical symptom combinations for all ten diseases. For example, the symptoms fever, cough, loss of taste and shortness of breath were correctly predicted as **COVID-19**, while headache, nausea and dizziness were predicted as **Migraine**.

11. Conclusion

This project successfully demonstrates a complete machine learning workflow — from data loading and preprocessing to model training, prediction and result display — for predicting diseases from symptoms. The Decision Tree Classifier provides accurate, interpretable predictions and the application handles user input robustly. It is a solid, beginner-friendly foundation for understanding applied machine learning.

12. Future Scope

- Add more diseases and symptoms for broader coverage.
- Train on real, large-scale clinical datasets.
- Compare with other models such as Random Forest, Naive Bayes and SVM.
- Build a web or mobile front-end for wider accessibility.
- Incorporate symptom severity levels instead of simple yes/no values.
- Integrate with electronic health records for personalised predictions.